Maximum Flows and Minimum Cuts

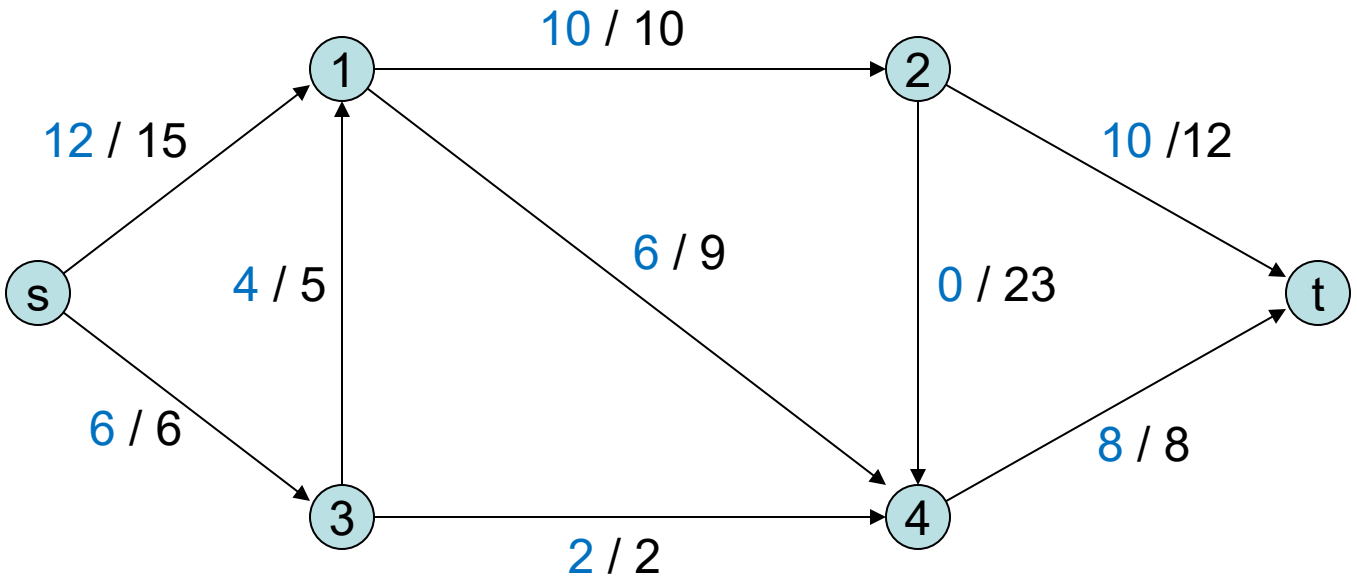
Design and Analysis of Algorithms

Brian C. Dean

School of Computing

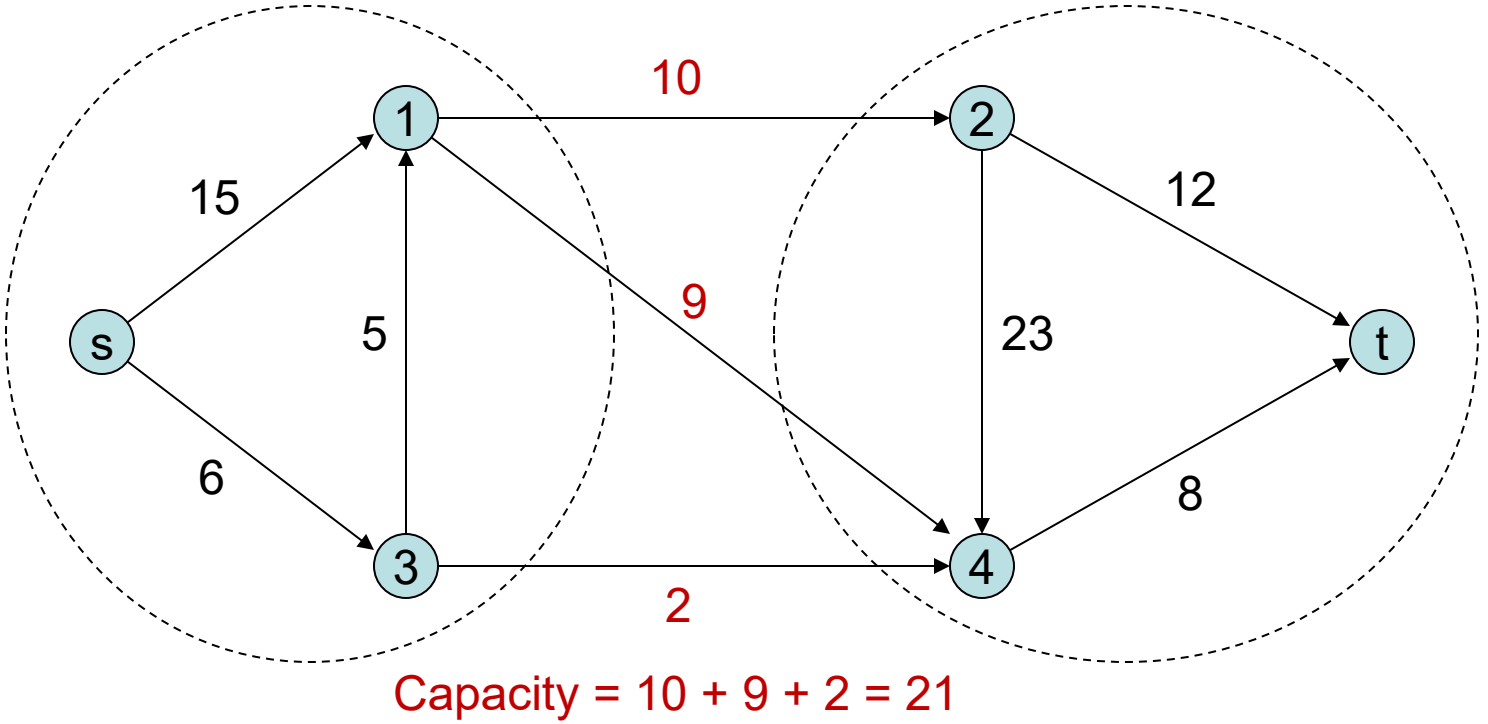
Clemson University

The Maximum Flow Problem



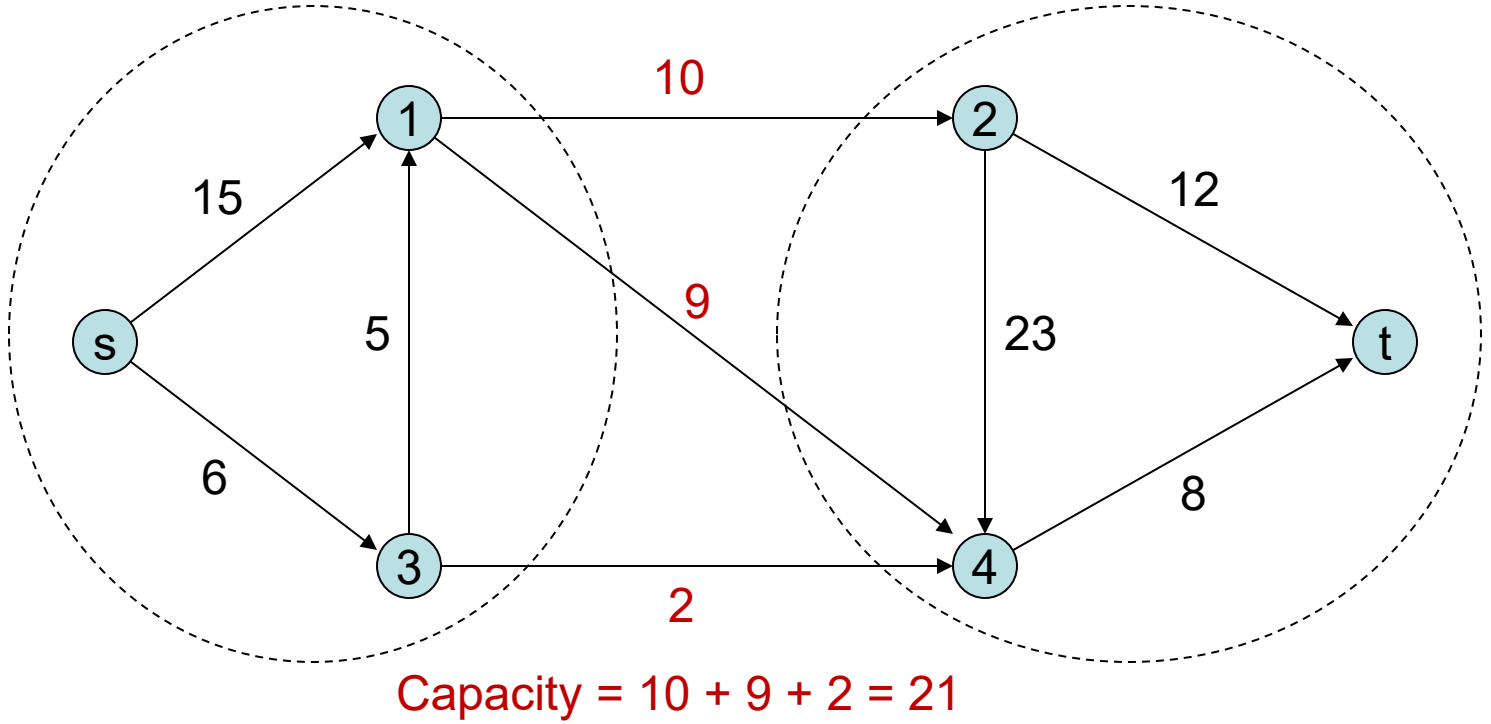
Given a directed graph with capacities on edges, what is the maximum amount of flow that can be shipped from a source node s to a sink node t? 18 units

s-t Cuts



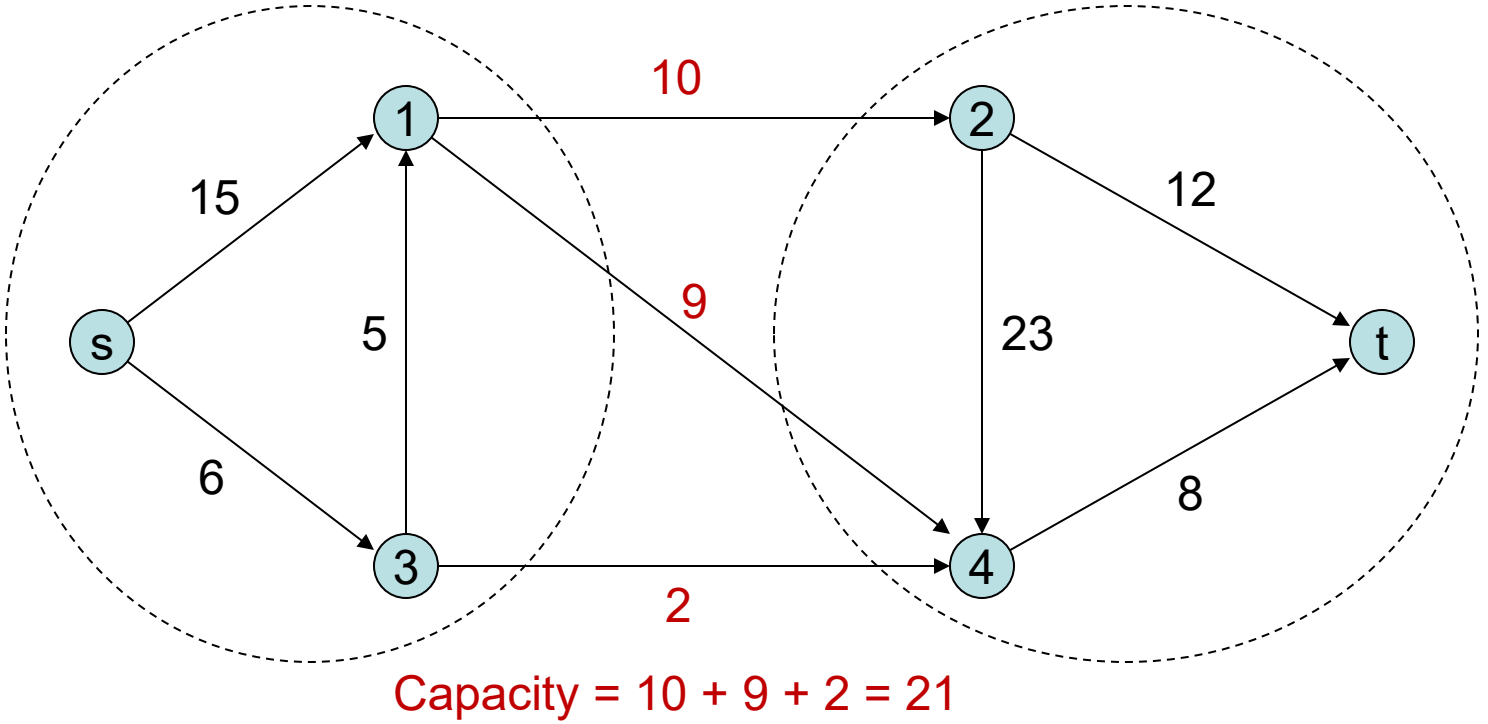
- An s-t cut is a partition of the nodes in our graph into two sets, one containing s and the other containing t.
- The capacity of a cut is the sum of the capacities of the edges crossing the cut in the s → t direction.

s-t Cuts



- **Easy observation:** The value of the maximum flow is at most the capacity of any cut.
- So: max s-t flow value ≤ min s-t cut capacity

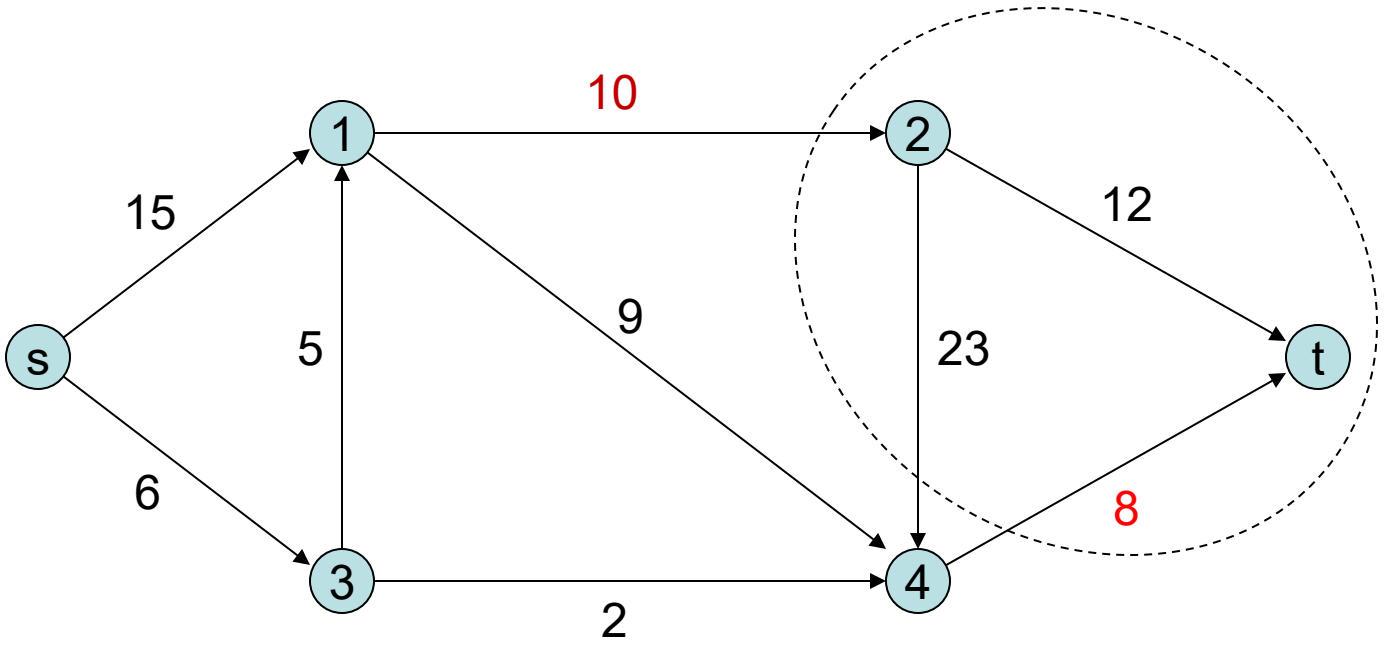
s-t Cuts



Capacity = 10 + 9 + 2 = 21

- **Easy observation:** The value of the maximum flow is at most the capacity of any cut.
- So: max s-t flow value ≤ min s-t cut capacity
- **Famous result:** max s-t flow value **=** min s-t cut capacity.

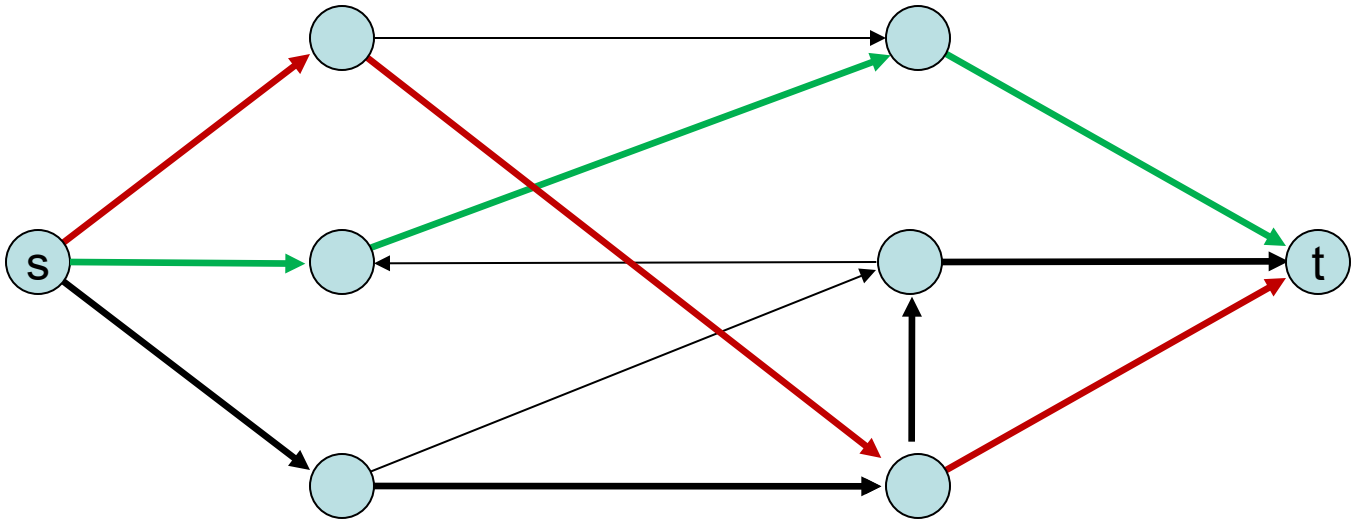
s-t Cuts



Capacity = 10 + 8 = 18

- **Easy observation:** The value of the maximum flow is at most the capacity of any cut.
- So: max s-t flow value ≤ min s-t cut capacity
- **Famous result:** max s-t flow value **=** min s-t cut capacity.

Higher Levels of Connectivity



Maximum # of edge-disjoint s-t paths

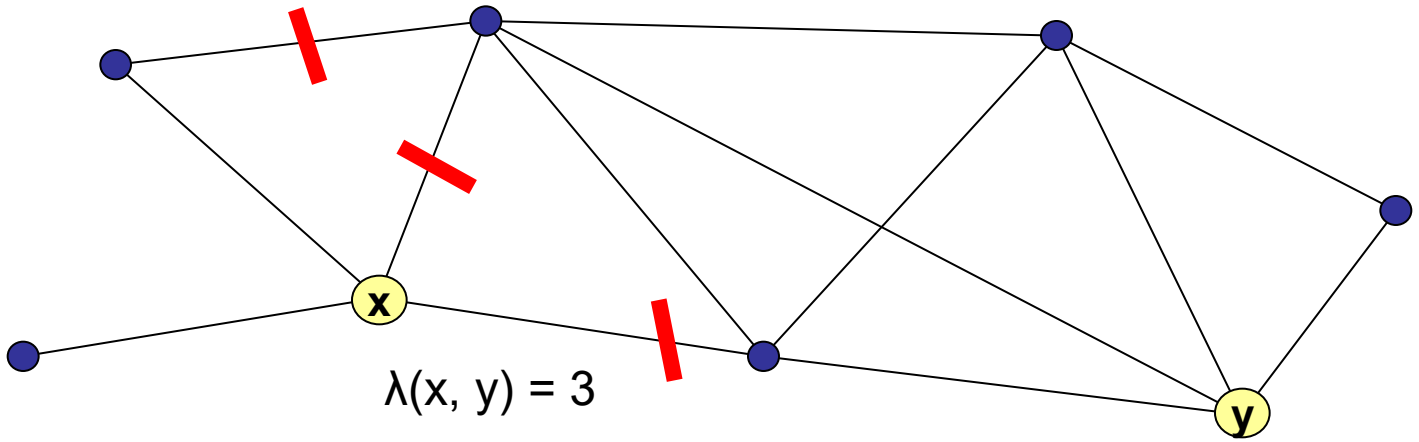
(path decomposition of a max flow in a graph with unit capacities)

=

Minimum # of edges we need to remove to separate s and t.

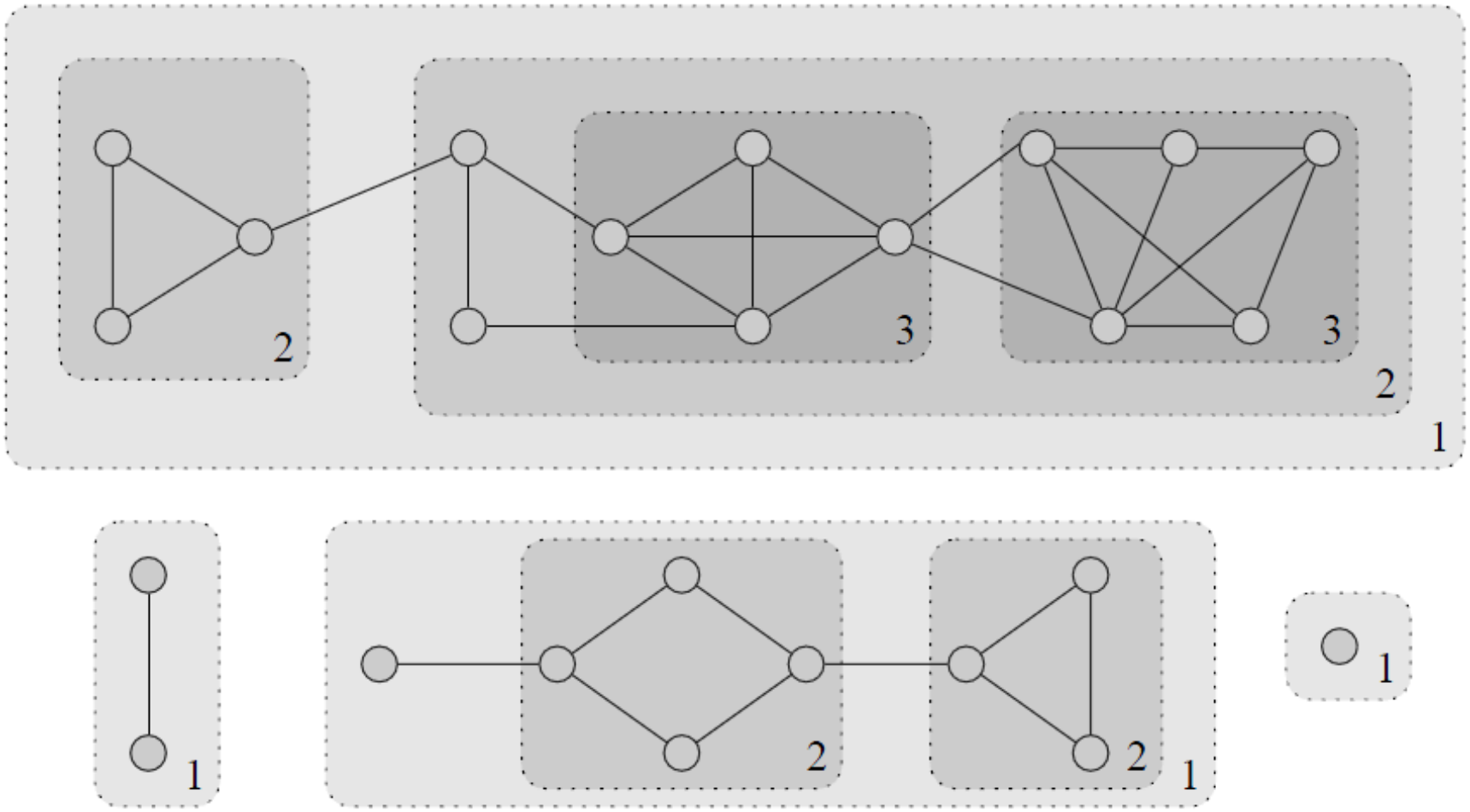
(minimum s-t cut)

Higher Levels of Connectivity



- Let λ(x,y) denote the value of a unit-cap max flow from x to y.
- Equivalently, the minimum # of edge removals to separate x from y.
- This quantity is known as the edge connectivity between x and y.

The Hierarchical, Nested Structure of k-Edge-Connected Components



Calculating Max Flows

- Ford-Fulkerson algorithm: repeatedly push flow through s-t “augmenting paths”.
- When no augmenting path available, we’ve found a matching s-t flow and s-t cut, both of which therefore are optimal!

Calculating Max Flows

- Ford-Fulkerson algorithm: repeatedly push flow through s-t "augmenting paths".
- Some possible running times:

Type of Aug Path:	Running Time per Augmentation	# Augmentations	Total Runtime
Arbitrary	O(m), via DFS	O(F) (F = max flow)	O(mF)
Shortest	O(m), via BFS	O(mn)	$O(m^2n)$
Fattest	O(m + n log n), via Dijkstra	O(m log F)	O((m + n log n)(m log F))

(in practice, things usually run much faster than running times would predict)